# Forensics for Cybercrimes Committed Using Artificial Intelligence

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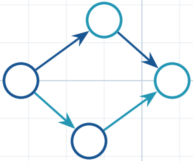
2026

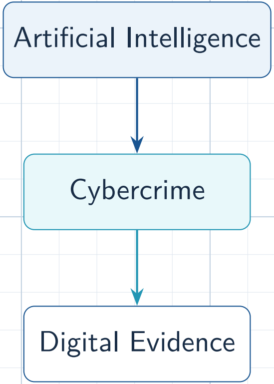


# Research Context

## The Emerging Problem

Artificial intelligence is increasingly being used to:

- automate cyberattacks;
- generate deceptive digital content;
- impersonate individuals and organisations;
- manipulate digital evidence;
- scale traditional cybercrime.



# The Forensic Challenge

**Traditional digital forensics**

- Identify the device
- Acquire the evidence
- Preserve integrity
- Analyse artefacts
- Establish attribution

**AI-enabled cybercrime**

- Synthetic identities
- AI-generated content
- Automated attacks
- Adversarial manipulation
- Difficult attribution

### Research Question

How can digital forensic methods reliably identify, preserve, analyse and attribute evidence arising from cybercrimes committed using artificial intelligence?

# Formalising Digital Evidence

Let the set of digital evidence be represented by

$$E = \{e_1, e_2, \ldots, e_n\}$$

where each element represents a digital artefact.
For example:

$$E = \{\text{email}, \text{metadata}, \text{log}, \text{image}, \text{network packet}\}$$

We can define a forensic evidence function:

$$F(e_i) = \{A_i, I_i, P_i, T_i\}$$

where:

- $A_i$ = authenticity
- $I_i$ = integrity
- $P_i$ = provenance

# Evidence Relationships

Digital evidence does not exist in isolation.
Consider:

$$G = (V, E)$$

where:

- $V$ represents digital entities;
- $E$ represents relationships between entities.

Examples of nodes:

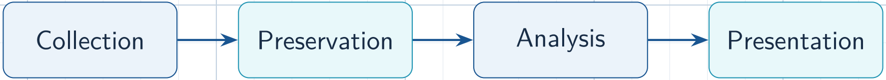
$$V = \{\text{user}, \text{device}, \text{account}, \text{IP}, \text{email}\}$$

Examples of edges:

$$E = \{\text{login}, \text{communication}, \text{transfer}, \text{execution}\}$$

Forensic implication

# Digital Forensic Evidence Pipeline




1. **Collection** — acquire relevant digital artefacts.
2. **Preservation** — protect integrity and provenance.
3. **Analysis** — reconstruct events and relationships.
4. **Presentation** — communicate findings for decision-making.

# Research Objectives

1. Examine existing digital forensic approaches for AI-enabled cybercrime.
2. Identify forensic challenges introduced by artificial intelligence.
3. Develop a framework for collection, preservation, analysis and presentation of AI-related digital evidence.
4. Investigate methods for authenticity, provenance and attribution.
5. Validate the proposed framework using appropriate forensic scenarios and datasets.

# Expected Contribution

### Theoretical

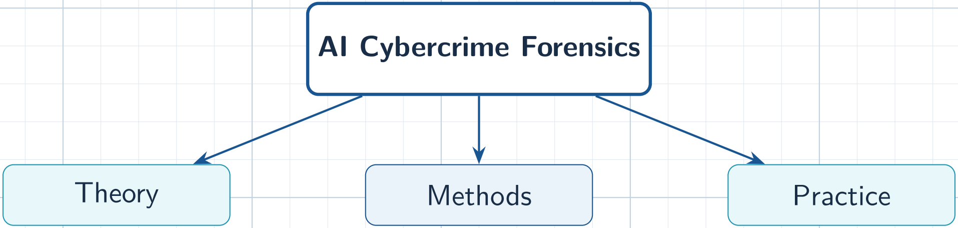
Formalisation of forensic reasoning for AI-enabled cybercrime.

### Technical

Methods for analysing authenticity, provenance and attribution.

### Practical

A forensic framework applicable to real-world investigations.



# Why This Matters

## The central argument

As cybercrime evolves from human-operated attacks toward AI-assisted and AI-generated attacks, forensic science must also evolve.

$$\boxed{\text{AI-enabled crime} \quad \Longrightarrow \quad \text{AI-aware forensic reasoning}}$$

The research therefore seeks to bridge:

**Artificial Intelligence** $\leftrightarrow$ **Cybersecurity** $\leftrightarrow$ **Digital Forensics**

# Conclusion

- AI is changing the nature and scale of cybercrime.
- Digital evidence associated with AI-enabled crime requires stronger methods of authentication and attribution.
- Formal reasoning, graph analysis and forensic methodologies can complement conventional investigations.
- The proposed research aims to develop a defensible framework for AI-enabled cybercrime forensics.

**From digital artefacts to defensible forensic conclusions**

# Thank You

## Questions and Discussion

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